Q9. Domain, host and website information

Original question: Gather publicly available information about an authorized domain or website, including domain details, host information, and web technologies. Interpret the information collected.

Aim

Map an approved domain to DNS/hosting clues and website technology information, and distinguish registration, DNS and serving infrastructure.

Requirements

Authorized domain/URL; whois, dig and WhatWeb. This extends Q2 by explicitly collecting host records and explaining CDN/shared-host limitations. References: [R1], [R2].

Procedure

Step 1 - Registration and host records

```
mkdir -p "$HOME/EHF_Exam/Q9"
cd "$HOME/EHF_Exam/Q9"
DOMAIN='<AUTHORIZED_DOMAIN>'
URL='https://<AUTHORIZED_DOMAIN>/'
date -Is | tee collected_at.txt
whois "$DOMAIN" > registration.txt
dig "$DOMAIN" A > host_A.txt
dig "$DOMAIN" AAAA > host_AAAA.txt
dig "$DOMAIN" CNAME > host_CNAME.txt
dig "$DOMAIN" NS > host_NS.txt
cat registration.txt host_A.txt host_AAAA.txt host_CNAME.txt host_NS.txt
```

A records return IPv4 addresses; AAAA returns IPv6; CNAME may identify an alias; NS identifies DNS authority. Some apex services flatten aliases so no CNAME is visible. A result containing multiple addresses may reflect balancing or a CDN. Keep resolver and TTL information from the full responses.

Step 2 - Inspect reverse DNS for an actual returned address

```
HOST_IP='<OBSERVED_DNS_IP>'
dig -x "$HOST_IP" > reverse_dns.txt
cat reverse_dns.txt
```

Use only an address actually returned in Step 1. PTR records, if present, can suggest a provider's naming convention; they do not prove ownership, location or that the server belongs exclusively to the domain. NXDOMAIN/no PTR is a valid result. Do not port-scan these addresses: the domain's approved website scope does not automatically authorize shared infrastructure scans.

Step 3 - Identify web clues and save records

```
whatweb -a 1 --log-verbose=web_technology.txt "$URL"
sed -n '1,100p' web_technology.txt
ls -l
```

Record HTTP status, redirect destination, visible server/framework clues and collection time. Use only in-scope redirects as in Q2. Treat WAF/CDN fingerprints as evidence about the serving layer; they need not reveal the origin server or application version.

Expected output - illustrative mapping

```
Registration provider: Example Registrar
A answer: 192.0.2.20
PTR: edge20.example.net.
Web clue: reverse-proxy/server header present
```

These are synthetic values using reserved example names/addresses. They are not a discovery result.

Interpretation, observation and result

Make a table: registration provider | authoritative DNS | DNS address/alias | PTR clue | web clue | confidence/limit. A registrar administers registration, an NS set provides DNS authority, and an A/AAAA record points to a current serving address. A CDN or shared host may serve many unrelated sites; an IP does not identify the origin or the domain owner. Report each clue at the layer it supports. The practical is complete when host information has been explicitly collected and related to, rather than confused with, registration and website fingerprints.

Common errors

- CNAME absent: an A/AAAA answer may still be valid; don't invent an alias.
- Different DNS answers between runs: record time, resolver, TTL and the changed answer; balancing can be normal.
- No technology version: record unknown rather than guessing from page appearance.

Viva questions

- What does a PTR record do? It maps an IP address to a reverse-DNS name when configured.
- Can one IP host many websites? Yes; shared hosting, proxies and CDNs commonly do.
- Is an A record necessarily the origin server? No; it may be an edge or proxy address.
- How is Q9 broader than Q2? It explicitly maps DNS/host clues and separates them from registration and web clues.